

Energy Systems - Group Project

Group O

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1 Task 1 - Questions on primal solution

(a) Compute the annuity factor $a(r, L)$ and the annualised costs for the technologies.

Output: Table (0.5)

With $r = 7\%$ and the following equations for the annuity factor $a(r, L)$ and the annualised costs c_{inv}

$$a(r, L) = \frac{r(1+r)^L}{(1+r)^L - 1} \quad [-]$$

$$c_{inv} = \text{overnight} \cdot a(r, L) + \text{FOM} \quad \left[\frac{\text{EUR}}{\text{MW(h)} \cdot \text{a}} \right]$$

we receive these values for the different technologies:

Component	Kind	$c_{\text{var, other}}$ [$\frac{\text{EUR}}{\text{MWh}}$]	Overnight Cost [$\frac{\text{EUR}}{\text{MW(h)}}$]	FOM [$\frac{\text{EUR}}{\text{MW(h)} \cdot \text{a}}$]	Lifetime [a]	Efficiency [-]	Annuity Factor [-]	c_{inv} [$\frac{\text{EUR}}{\text{MW(h)} \cdot \text{a}}$]
Wind	power	1.62	1,286,467	15,148	30	–	0.08	118,819.75
Solar	power	0.00	367,867	9,304	40	–	0.08	36,897.39
H ₂ electrolyzer	charge power	0.00	1,257,335	50,293	25	0.70	0.09	158,185.57
H ₂ fuel cell	discharge power	0.00	1,068,642	53,432	10	0.50	0.14	205,582.58
H ₂ storage	energy	0.00	1,604	0	100	–	0.07	112.41
Battery inverter	power	0.00	80,223	722	10	0.96	0.14	12,143.95
Battery storage	energy	0.00	100,279	0	30	–	0.08	8,081.12
Line	power	0.00	1,544,274	23,164	40	–	0.08	138,998.66

Table 1: Annuity factor and annualised costs for the technologies. MW(h) denotes MW for power components and MWh for energy components.

(b) In the no-line case, what does the dispatch of solar and wind look like in each node?

Output: Dispatch plot (0.5)

To find the optimal solution in the no-line case, we minimize the objective function subject to the constraints and receive with the dual-simplex algorithm the value of

$$\min z = 3\,412\,846\,097.4 \text{ EUR}$$

It can be observed that wind and solar are combined only in the northern node, whereas the southern node instead uses a combination of solar power and storage. This is due to several reasons. In scenario B (no line), each node has to cover its own demand, so the dispatch follows the local resource.

Generation dispatch:

In the north, the supply is a mix of wind and solar. Wind carries the base supply across day and night and dominates in winter, while solar adds generation during the daytime hours, so the dispatch stays relatively continuous. In the south, no wind is built at all, so solar is the only source. Its dispatch is strongly diurnal: high peaks around midday and zero at night, with large swings between the seasons.

Storage dispatch:

Because the north already has a fairly balanced wind-solar mix, it needs less storage. The south depends entirely on solar, which produces nothing at night, so a lot of energy has to be shifted from day to night. That is why both the battery and the H₂ charge/discharge power and storage capacity are clearly higher in the south. Batteries handle the short day-night cycle, while H₂ covers the longer, seasonal shifts. Overall, this split is the solver's least-cost choice: it follows the local production potential (north = windiest node, south = sunniest node) together with the technology costs.

(c) Compare the two scenarios. How do the investments into generation, storage, and transmission change?

Output: Show the investment costs in a table and plot the capacities.(1)

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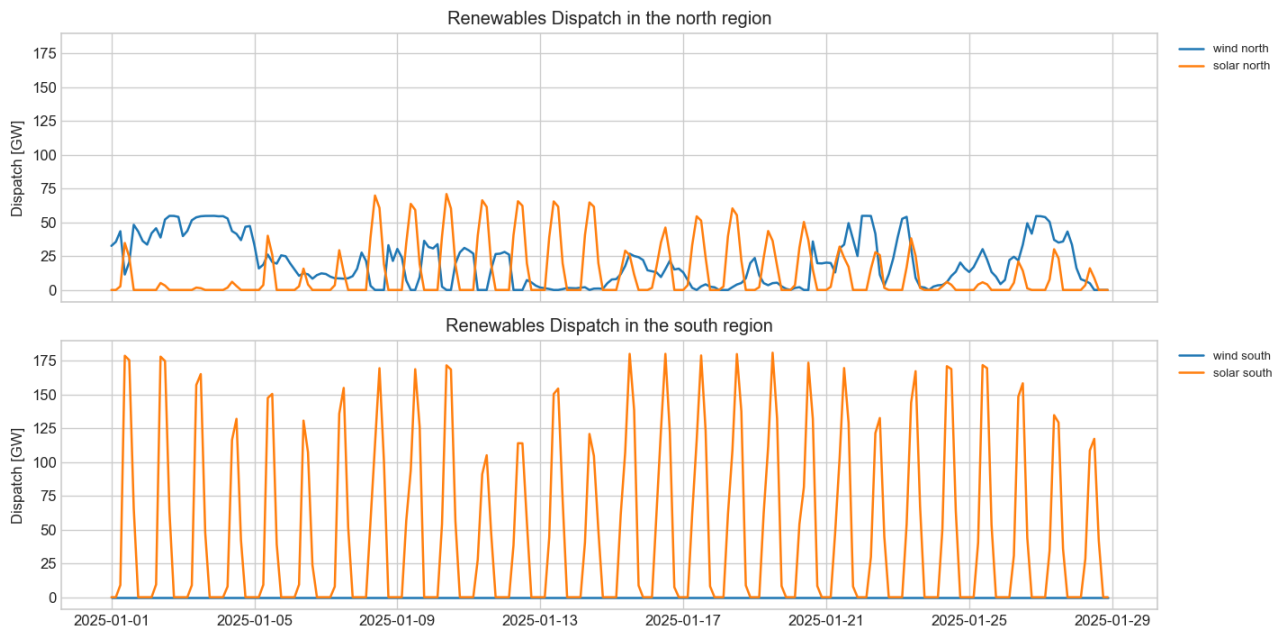


Figure 1: Dispatch of solar and wind in each node in the no-line case.

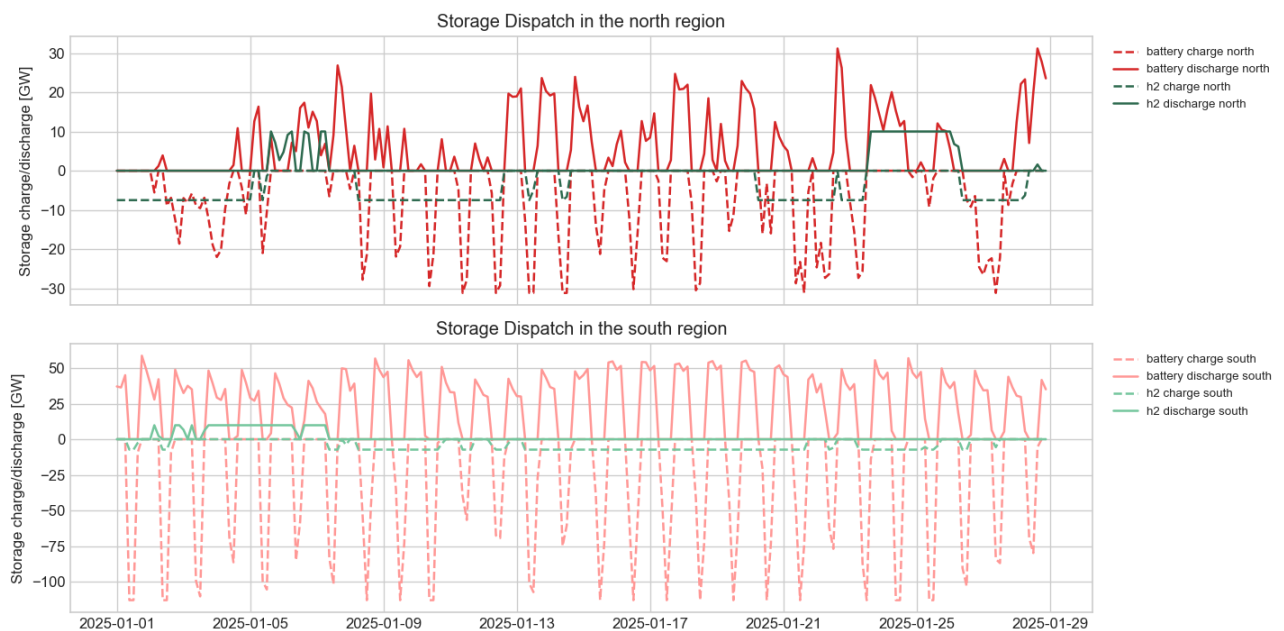


Figure 2: Participation of storage systems in the dispatch pattern in each node.

TOTEX Scenario A (line)	3 056 740 408.5 EUR
TOTEX Scenario B (no line)	3 412 846 097.4 EUR
Savings through line inclusion	356 105 688.9 EUR
Build line capacity	14.06 GW

Table 2: Optimal solution for scenarios A and B – investment cost and built line capacity.

Component	A (line) [$\frac{\text{MioEUR}}{\text{a}}$]	B (no line) [$\frac{\text{MioEUR}}{\text{a}}$]	Δ A-B [$\frac{\text{MioEUR}}{\text{a}}$]
Wind	8 385	6 858	1 527
Solar	15 714	19 105	-3 391
H ₂ electrolyzer	859	2 343	-1 484
H ₂ fuel cell	1 824	4 078	-2 254
H ₂ storage	167	359	-192
Battery inverter	1 536	1 751	-215
Battery storage	9 085	9 714	-629
Line	1 954	0	1 954

Table 3: Annual costs comparison between scenarios A and B for every technology.

The following Table 2 shows the optimal solution for scenarios A and B, as well as savings through line inclusion (which is the cost of B - A) and the built line capacity in scenario A.

Figure 3 shows the comparison of the power and energy capacity in each node and for both scenarios.

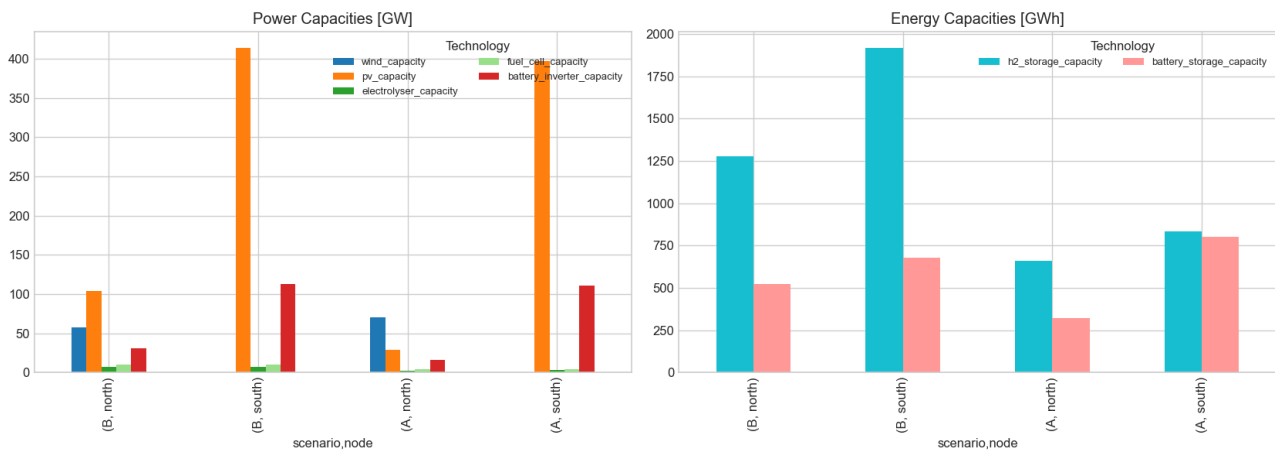


Figure 3: Power and energy capacity for each node and scenario.

The total investment costs shrink when line capacity is built (TOTEX falls from 3.41 to 3.06 billion EUR). With the line, the regions no longer have to cover their demand on their own, so they can specialise.

Generation:

The north expands wind and strongly cuts its solar, turning into a cheap wind exporter, while the south stays solar-based. Wind in the north is favourable because its good resource potential combined with the costs helps to reduce total costs. The south keeps its strong solar potential and only needs to reduce its storage and a little of its production capacity to meet demand.

Storage:

Because excess energy can be transferred between regions instead of being stored locally, less storage is needed overall. This is mostly visible in the H₂ storage (it drops by roughly half) and in the northern battery, while the southern battery even grows slightly.

Transmission:

Line capacity is built to carry the energy and to smooth the supply across seasonal and day/night differences between the windy north and the sunny south.

(d) How does curtailment by technology and region change from the no-line case to the transmission

case and why?

Output: Quantify and explain. (0.5)

The curtailment of wind and solar generation in the respective regions is shown in Figure 4 and Figure 5. Figure 4 illustrates the curtailment over time for each scenario, while Figure 5 presents the corresponding values aggregated over time and disaggregated by technology and node. Table 4 provides the numerical values underlying the visualization in Figure 5.

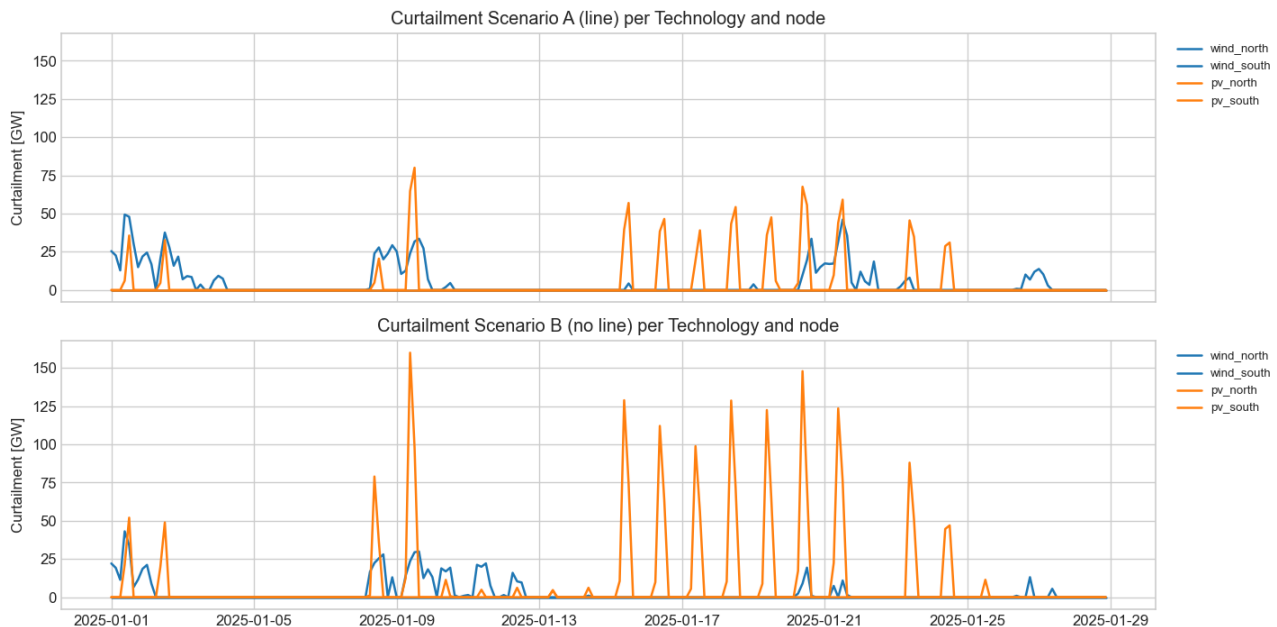


Figure 4: Time series of curtailment in each scenario.

Technology and Region	Scenario A [GWh]	Scenario B [GWh]
Wind north	2 044	3 376
Wind south	0	0
Solar north	66	0
Solar south	6 641	3 169

Table 4: Curtailment in GWh for each technology, region and scenario.

Total curtailment falls when the line is built, but it does not fall everywhere.

South PV:

Curtailment drops strongly. In the no-line case, the south has a large midday solar surplus that it cannot use or store, so it has to be spilled. With the line, this surplus can be exported to the north instead of being curtailed.

North wind:

Curtailment actually increases. With the line, the north overbuilds cheap wind to export it. During high-wind hours, the line and storage cannot always absorb all of it, so a larger share of the wind potential is spilled.

North PV:

It drops to zero because the north barely builds any solar anymore in the line scenario. So the line reduces overall curtailment by letting the regions share their surplus, but it shifts the remaining curtailment towards the technology that is deliberately overbuilt for export (northern wind).

(e) How do the residual load curves change across the two scenarios?

Output: Plot the residual load curves and explain the differences. (0.5)

Figure 6 illustrates the residual load curve for the line and no-line cases.

The residual load is defined as the difference between electricity demand and dispatched renewable generation, aggregated over both nodes and sorted into a duration curve:

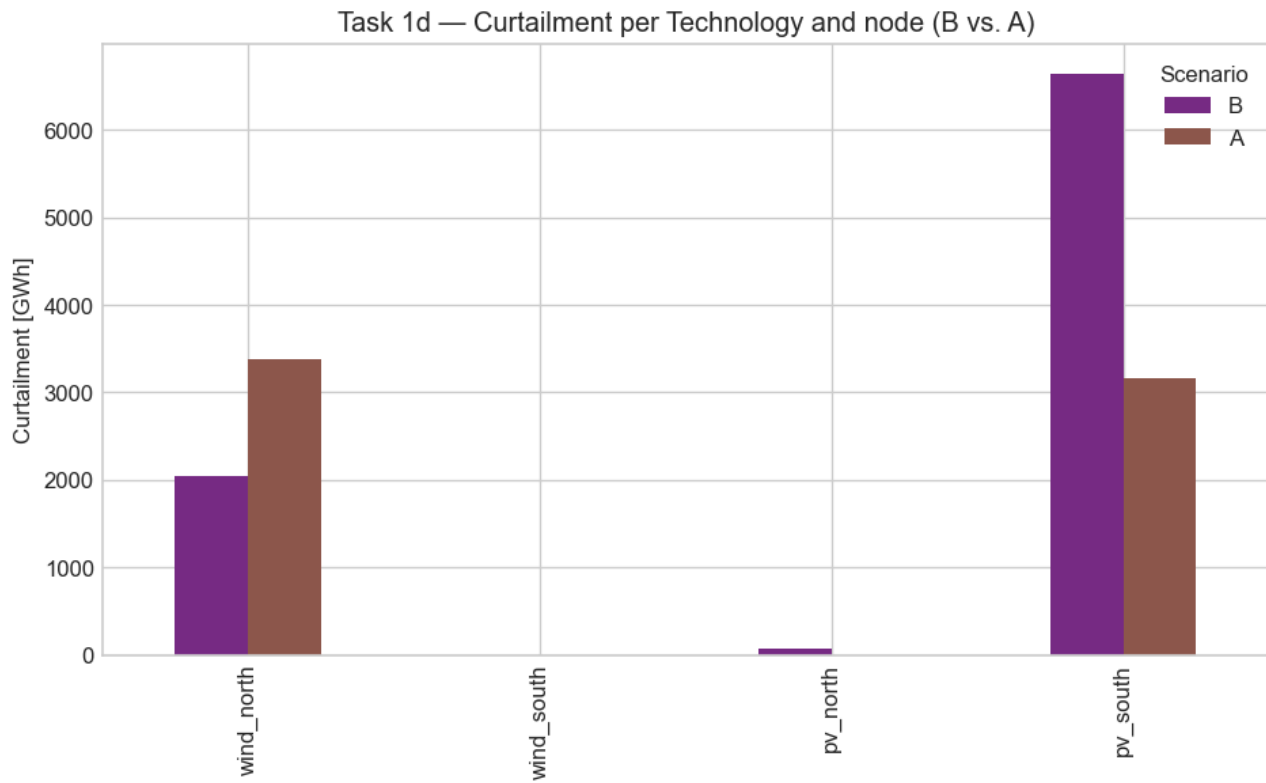


Figure 5: Curtailment in GWh for each technology, region and scenario.

$$R(t) = D(t) - G_{\text{RES}}(t) \quad [\text{MW}]$$

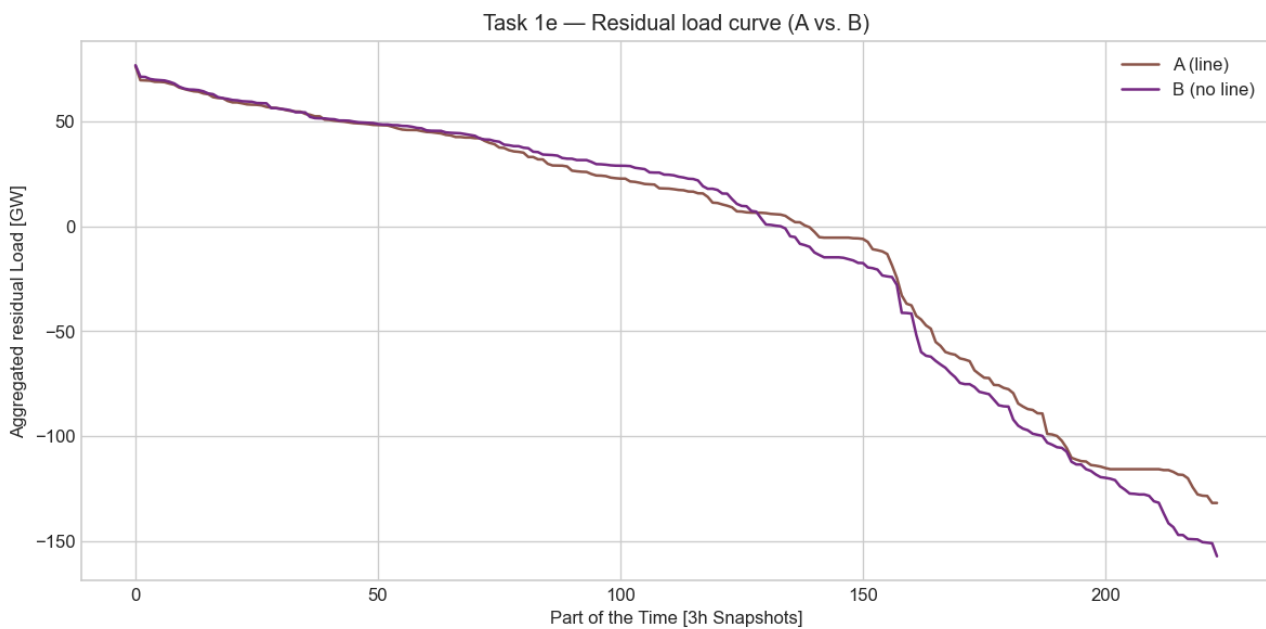


Figure 6: Residual load curve of scenarios A and B.

The peak of the curve is almost identical in both scenarios. This corresponds to the system-wide situation in

which neither wind nor PV generation is available. When both regions simultaneously experience low wind and solar availability, the transmission line cannot provide additional support, since no surplus energy is available for spatial balancing. Consequently, storage must cover the demand. The main difference appears at the negative (surplus) end of the curve: with the transmission line, the curve is significantly flatter. The line enables spatial balancing of non-simultaneous surpluses between regions (northern wind and southern solar), thereby reducing the amount of energy that needs to be shifted through storage. This is consistent with the results in Sections 1c and 1d: with transmission, the system relies less on storage (resulting in lower hydrogen capacities) and experiences less curtailment, as spatial exchange partially substitutes temporal storage.

2 Task 2 - Questions on dual solution

(a) For each scenario, extract the shadow variable to the nodal energy balances and weight it by the nodal demand in the north and south, respectively. What do they represent? Why are they different in the two regions? When do you observe the highest values and why? (0.5)

The nodal electricity price at each node n and time step t is derived from the dual variable $\mu_n(t)$ of the power balance constraint, divided by the time resolution $\Delta t = 3 \text{ h}$:

$$\lambda_n(t) = \frac{\mu_n(t)}{\Delta t} \quad \left[\frac{\text{EUR}}{\text{MWh}} \right]$$

The demand-weighted average price for each region is then computed as:

$$\bar{\lambda}_n = \frac{\sum_t \lambda_n(t) \cdot d_n(t)}{\sum_t d_n(t)}$$

Figure 7 shows the sorted nodal price duration curves for both scenarios and both regions.

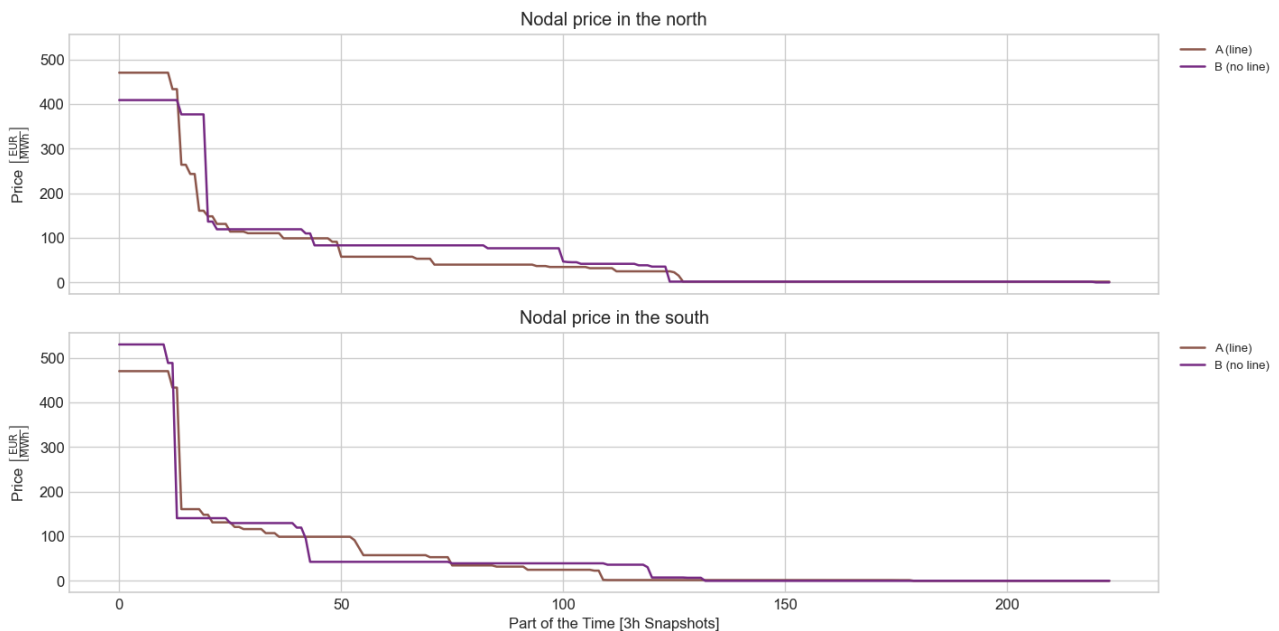


Figure 7: Nodal price duration curves for scenarios A (line) and B (no line) in each region.

The price duration curve shows the nodal price for each share of the time, sorted from high to low. For example, in roughly the most expensive fifth of the hours the price is above $100 \frac{\text{EUR}}{\text{MWh}}$. The dual variable of the nodal

energy balance represents the marginal cost of supplying one additional MWh at the respective node, i.e. the local electricity price ($\frac{\text{EUR}}{\text{MWh}}$). Weighting it by the nodal demand gives the average price paid by consumers in that region over the modelled period.

In scenario B, prices differ because each node forms its own price independently. The south has large near-zero-cost midday solar surpluses that pull its average down, while the north's wind-dominated mix has almost no surplus hours, so wind's variable cost ($1.62 \frac{\text{EUR}}{\text{MWh}}$) sets the price most of the time, resulting in a higher average. In scenario A, the line couples the two prices during hours of full capacity, flattening both duration curves and bringing regional averages closer together.

The highest price values are observed during dark-lull hours when neither wind nor solar generation is available. In these scarcity periods the only remaining dispatchable source is the H_2 fuel cell, whose dispatch sets the marginal price. In scenario B the extreme price peaks occur in the south, as a solar-only region it is most exposed when there is no sunshine. In scenario A the transmission line couples both nodes during scarcity, so both regions reach the same peak price simultaneously (the southern extreme is lowered and the northern peak is raised until they converge).

(b) In the transmission scenario, consumers of which region benefit more from the transmission line: the north, the south, or both equally? Briefly explain. (0.5)

Figure 8 compares the demand-weighted average nodal prices for both regions across the two scenarios.

Consumers in the north benefit more from the transmission line, although both regions gain. Northern consumers benefit more. In scenario B the north had the higher average price due to its tight wind-solar mix with almost no zero-price hours. The line gives the north access to cheap southern solar surplus, avoiding additional PV and storage investment and producing a larger absolute price reduction. The south benefits too, as it can export surplus instead of curtailing it and needs less storage, but its price reduction is smaller.

The line thus balances the two nodal prices and reduces them in both regions, but to a greater extent in the north.

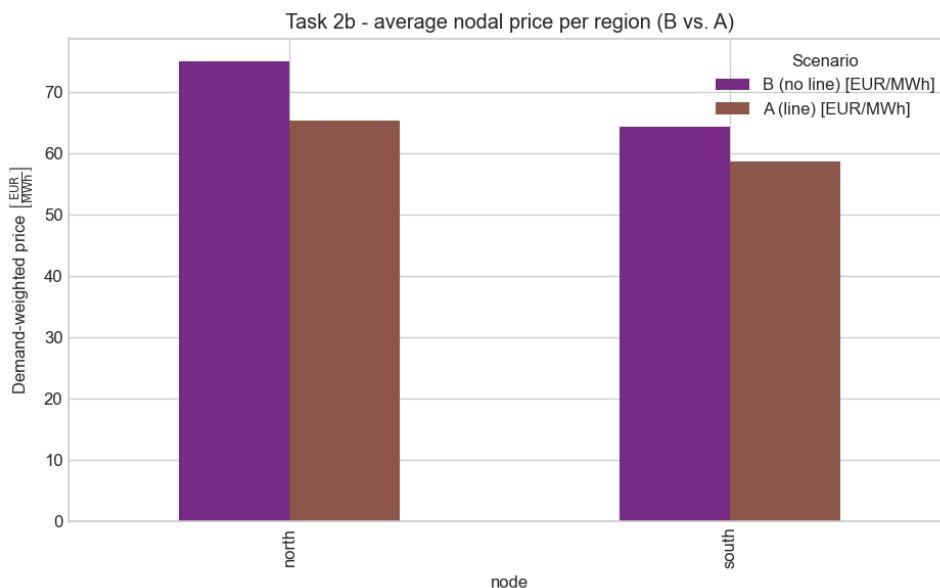


Figure 8: Demand-weighted average nodal price per region in scenarios A (line) and B (no line).

(c) Extract the shadow price of the line capacity constraint in the transmission scenario. What does it represent? In the scenario where the transmission capacity is zero, you can also extract the dual variable. Comparing the two, what does it say about the value of interconnection? (0.5)

The shadow price of the line capacity constraint is obtained by summing the duals of the upper and lower transmission flow constraints over all time steps. Figure 9 illustrates the comparison between the two scenarios.

The shadow price represents the congestion rent: how much the total system cost would drop if the line could carry one more MW. It corresponds to the price difference between north and south in the hours when the line

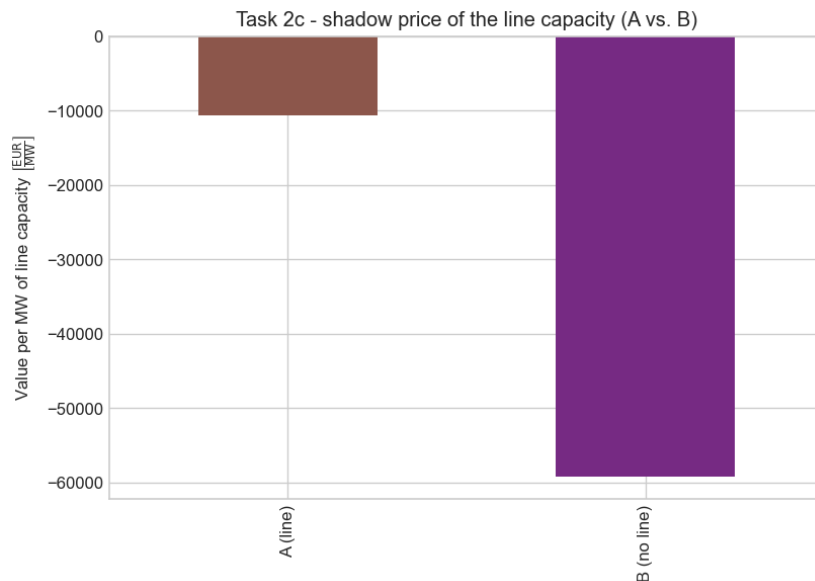


Figure 9: Shadow price of the line capacity constraint in scenarios A (line) and B (no line).

is operating at its capacity limit.

In scenario A (line), the shadow price is small (about $-11 \frac{\text{kEUR}}{\text{MW}}$). Since the model is free to optimise the line capacity, it expands the line until the marginal benefit of one additional MW equals its annualised investment cost, so the residual marginal value is close to zero by construction.

In scenario B (no line), the line capacity is forced to zero. The no-line constraint reflects the value of the very first MW of interconnection (about $-59 \frac{\text{kEUR}}{\text{MW}}$). This value is roughly 5.6 times larger than the marginal value in scenario A, illustrating strongly decreasing returns to interconnection. Interconnection is highly valuable, but with clearly decreasing value. The first connections yield the greatest cost savings and additional capacity is built only as long as its marginal benefit exceeds its marginal cost.

(d) Which advantages and disadvantages of heavy reliance on cross-regional interconnection does this stylised model not capture? No calculation needed. (0.5)

The model captures the cost minimising value of interconnection under idealised conditions, but neglects several important real world effects.

Disadvantages not captured:

Highly interconnected systems are susceptible to cascade failures and systemic risk not representable in a lossless single-line model. Cross-border operation requires regulatory coordination and is exposed to political risk such as unilateral export restrictions. Large transmission projects also face public acceptance barriers that can delay or prevent construction.

Advantages not captured:

Beyond direct cost savings, interconnection improves system security by enabling imports during local shortfalls, reduces local market power, and promotes price convergence. The two-node topology also ignores meshed grid benefits such as N-1 security and reactive power support, so the model understates the true social value of interconnection. Overall, the model provides a partial view of the true social value of interconnection, since it omits both the non-market benefits and the systemic risks that become relevant at high integration levels.